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Magnetic, physical, and lithologic properties and age of till exposed along the east coast[1] of Lake Huron, Ontario

C. P. GRAVENOR AND M. STUPAVSKY

Department of Geology, University of Windsor, Windsor, Ont., Canada N9B 3P4

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Oriented blocks of till from 31 sites were obtained from the Late Wisconsin St. Joseph Till over a distance of about 208 km along the east coast of Lake Huron. An analysis of paleomagnetic remanence data shows that two units within the St. Joseph Till were sampled, although it has not been possible to find a consistent stratigraphic break from field examinations. These two units give separate virtual geomagnetic pole (VGP) positions of lat. 67.1° N, long. 297.8° W and lat. 65.1° N, long. 220.6° W. Remanence data from varved clays for the Walkerton area, which were deposited at about the same time as the St. Joseph Till, gave a VGP position close to the mean of all remanence results from the St. Joseph Till. A spruce log extracted from the till was dated at 13 100 ± 110 y B.P. (GSC-2213), which is consistent with the time of the Port Huron ice advance.

Granulometric analysis, carbonate analysis, and pebble analysis showed variations which are directly related to the nature of the underlying bedrock and could not be used to differentiate the two till units. Ratios of purple (dark red) to white (light red) garnets also demonstrate the presence of two till units within the St. Joseph Till and thus support the interpretation of the remanence data.

The results show that in the absence of direct field evidence remanence data is a sensitive tool which can be used to determine if more than one till unit is present.

Des blocs orientés de till provenant de 31 sites ont été prélevés dans le till de St-Joseph datant de la fin du Wisconsin; l'échantillonnage couvre une distance de 208 km le long de la côte est du Lac Huron. L'analyse de la rémanence paléomagnétique montre qu'il existe deux unités distinctes dans le till de St-Joseph quoiqu'il ne soit pas possible de situer une limite stratigraphique à partir d'observations sur le terrain. Ces deux unités indiquent des pôles géomagnétiques virtuels (PGV) distincts donts les coordonnées sont: 67.1° N et 297.8° O, 65.1° N et 220.6° O. Les données de rémanence pour les argiles varvées de la région de Walkerton, formées à peu près en même temps que le till St-Joseph, indiquent une position du PGV proche de la moyenne de tous les résultats obtenus pour le till. Un tronc d'épinette extrait du till donne un âge de 13 100 ± 110 ans (GSC-2213) ce qui est contemporain de la poussée glaciaire de Port Huron.

L'analyse granulométrique, la détermination des carbonates, et l'étude des galets montrent des variations qui se rattachent directement à la nature du roc sous-jacent et ne peuvent donc pas être utilisées pour différencier les deux unités de till. Les rapports entre grenats pourpres (rouge foncé) et grenats blancs (rouge pâle) aussi font supposer la présence de deux unités de till et ainsi supportent l'interprétation des données de rémanence.

Ces résultats montrent qu'en l'absence d'évidence sur le terrain, la rémanence est un outil sensible qu'on peut utiliser pour déterminer si plus d'une unité de till est présente.

[Traduit par le journal]

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[1]The word coast is used in this paper to mean the cliffs exposed by wave action on coastal region.

Introduction

Paleomagnetic studies made on the early Wisconsin Sunnybrook Till and the late Wisconsin Port Stanley and Leaside Till sheets in southern Ontario have confirmed that these units have a stable characteristic magnetization and can be characterized by their virtual geomagnetic pole (VGP) positions (Gravenor *et al.* 1973; Stupavsky *et al.* 1974*a*,*b*). The intent of this study is to examine the paleomagnetic properties, the magnetic fabric, and the physical and lithologic characteristics of an upper clay till, also of late Wisconsin age, exposed in the wave-cut cliffs along the east coast of Lake Huron between Sarnia and Port Elgin covering a shore-line distance of some 208 km (Fig. 1).

From previous work on the tills exposed along the east coast of Lake Huron it is known that at least two till sheets are present: an upper clay till, the St. Joseph Till (Cooper and Clue 1974), which has been assigned in age to the Wyoming moraine, which is the inner moraine of the Port Huron moraine system in the area under study (Fig. 1), and a lower stoney or sandy till of unknown age, but tentatively assigned as a correlative of the Catfish Creek Till (Leverett and Taylor 1915; Dreimanis 1961; Dreimanis and Goldthwait 1973; Cowan *et al.* 1975). As the St. Joseph Till has been assigned to the Wyoming moraine (Cowan *et al.* 1975) it is likely that it represents deposits from the last ice to occupy the Lake Huron basin and is somewhat younger than the Port Stanley Till found in the Lake Erie basin (Dreimanis and Goldthwait 1973). Thus paleomagnetic data obtained from this upper clay till should provide additional information on the VGP position during late Wisconsin time in southern Ontario.

Following the techniques used in previous studies (Gravenor *et al.* 1973), oriented blocks of till were obtained from 31 sites along the east coast of Lake Huron (Fig. 1). Samples from each site were analyzed for remanence data, anisotropy of magnetic susceptibility, grain size, pebble lithology, and carbonate content. The heavy minerals were separated from samples from 15 sites and the ratio of purple (dark red) to white (light red) garnets was determined. Oriented blocks of varved clays from the Walkerton area (about 30 km east of Kincardine, Fig. 1) were collected and remanence directions determined. This study was undertaken as a check on the validity of the VGP position for the upper clay till along Lake Huron, as in all likelihood the varves were deposited close to the time of deposition of the upper clay till.

Sampling and Measurement of Remanence

An average of about 5 oriented blocks of till measuring approximately 10 × 10 × 15 cm were cut out with masons' chisels from the cliffs at each site in a vertical sequence at intervals ranging from 10 to 200 cm. An average of 5 specimens (right cylinders) measuring 2.5 × 2.5 cm were cored from each block. After measuring the natural remanent magnetization (NRM) of all specimens using a Schonstedt SSM-1A spinner magnetometer, one pilot specimen per site with typical NRM was step demagnetized in alternating field (AF) in 100 oersted (Oe) steps from 100 to 1000 Oe peak intensities, using a Schonstedt GSD-1 AF demagnetizer. Guided by the stepwise changes in remanence directions of the pilot specimen, an optimum demagnetizing intensity was selected, using an objective stability index method (Symons and Stupavsky 1974) in which the remaining specimens from the site were AF demagnetized and their residual remanence remeasured (Table 1).

Three factors reveal that a stable characteristic remanence is being isolated in the specimens. First, the mean coercivity determined by AF demagnetization of the 31 pilot specimens is a relatively high 450 Oe. Second, the remanence directions of specimens within each block form tight clusters. Third, the remanence directions moved away from the present Earth's magnetic field in the initial 100 to 500 Oe of AF demagnetization as viscous components (VRM) were removed. The directions then clustered at demagnetizing fields of 200 to 700 Oe, after which they scattered increasingly as anhysteretic remanence components were induced in the specimens by the demagnetizing process. By giving each mean block (site) remanence direction unit weight, the mean site (till sheet) remanence direction and associated statistical quantities (Fisher 1953) were then calculated (Table 1).

Discussion of Remanence Results

The sampling of the St. Joseph Till was started during the summer of 1974 when some 16 sites were sampled in the area north of Goderich (Fig. 1). A visual examination of the remanence results from this study showed the presence of

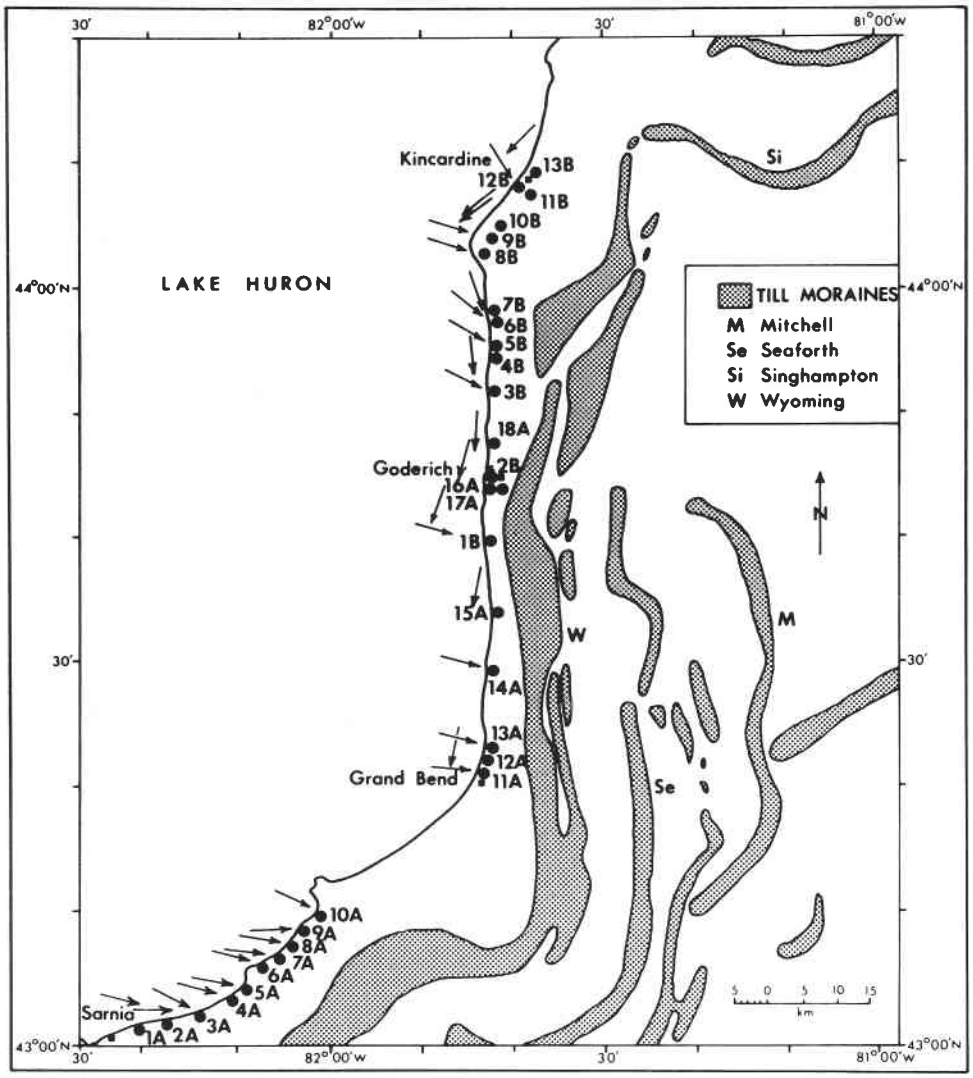


FIG. 1. Map showing the sampled locations from the east shore of Lake Huron and the position of the moraines associated with the Port Huron moraine system. The arrows indicate the direction of ice movement as determined from the mean maximum magnetic susceptibilities for each site. Position of moraines after Karrow (1974).

two distinct clusters. One of these remanence clusters was confined to the samples collected north of Goderich and the other south of Goderich. On the basis of this observation and the geographic separation of the sites it was suspected that two till sheets had been sampled. For purposes of discussion these tills have been informally designated the "A" till, which in general is found in the area south of Goderich, and the "B" till, which is found north of the Goderich area. In order to check this conclusion, in the summer of 1975 an additional 15 sites were sampled, 10 of which are located south of Grand Bend (Fig. 1) and the remainder taken from the area sampled in 1974.

These additional sites were assigned to the "A" or "B" till solely on the basis of geographic position. It is noted, however, from an inspection of Table 1 that several sites, 1A, 3A, 5A, and 10A, which have been assigned to the "A" till on the basis of location, belong to the "B" cluster, and site 12B probably belongs to the "A"

TABLE 1. Remanence data by site

Site	Number of specimens	Number of blocks	AF demagnetizing field (oersteds)	Length of resultant vector	Declination (degrees)	Inclination (degrees)	Precision parameter	α_{95} (degrees)
1A	18	3	700	2.96940	343.3	35.1	65	15.4
2A	17	3	500	2.95690	28.0	59.1	46	18.3
3A	24	4	700	3.91341	351.5	38.0	35	15.8
4A	24	4	700	3.99512	57.5	24.8	615	3.7
5A	36	6	700	5.90693	344.3	37.9	54	9.2
6A	46	8	600	7.95169	32.9	6.1	145	4.6
7A	24	4	600	3.99525	5.0	25.7	631	3.7
8A	24	4	600	3.83055	19.6	53.3	18	22.5
9A	24	4	600	3.98445	19.7	40.1	193	6.6
10A	24	6	600	5.93084	350.2	28.3	72	7.9
11A	28	6	600	5.82623	23.3	38.6	29	12.7
12A	22	5	600	4.97874	21.2	60.2	188	5.6
13A	19	4	200	3.93692	6.7	74.3	48	13.5
14A	33	6	600	5.48048	33.7	40.2	10	22.8
15A	10	2	600	1.94737	22.0	44.7	19	60.9
16A	22	5	600	4.37231	8.6	25.4	6	32.9
17A	36	5	200	4.96440	16.9	30.6	112	7.2
18A	16	3	300	2.97784	10.6	59.7	90	13.0
1B	22	5	500	4.97042	343.9	41.3	135	6.6
2B	38	6	600	5.90600	344.4	35.9	53	9.3
3B	14	3	200	2.92414	338.6	39.0	26	24.5
4B	18	4	700	3.96385	356.4	30.7	83	10.2
5B	20	5	700	4.64945	327.7	30.4	11	23.7
6B	30	6	600	5.73652	334.3	45.0	19	15.8
7B	30	5	600	4.75597	356.8	47.5	16	19.5
8B	24	4	600	3.95665	331.9	29.6	69	11.1
9B	19	3	600	2.98735	357.3	30.1	158	9.8
10B	31	5	400	4.95012	340.1	41.2	80	8.6
11B	24	4	600	3.98223	322.3	49.1	169	7.1
12B	23	4	500	3.94159	8.6	57.3	51	12.9
13B	23	4	500	3.83993	333.3	50.9	19	21.8
Mean								
A till	18	AF demagnetized		16.75453	14.4	41.3	14	9.7
A till	18	NRM		16.64299	12.5	56.5	13	10.2
B till	13	AF demagnetized		12.67586	342.4	41.4	37	6.9
B till	13	NRM		12.20006	341.9	57.0	15	11.1
A + B	31	AF demagnetized		28.80734	0.7	42.4	14	7.9
Walkerton varves	39	AF demagnetized		38.99227	3.2	35.8	38	3.8

Virtual Geomagnetic Poles

		Long. W	Lat. N	δ_p	δ_m
A till	AF demagnetized	297.8	67.1	7.2	11.8
A till	NRM	323.4	78.7	10.6	14.7
B till	AF demagnetized	220.6	65.1	5.1	8.4
B till	NRM	190.6	74.9	11.7	16.1
A + B	AF demagnetized	263.5	70.5	5.5	8.9
Walkerton varves	AF demagnetized	268.4	65.6	2.5	4.4

NOTES: α_{95} = radius of cone of 95% confidence.
$\delta_p\delta_m$ = semi-axes of oval of 95% confidence along and perpendicular to the site-pole great circle.

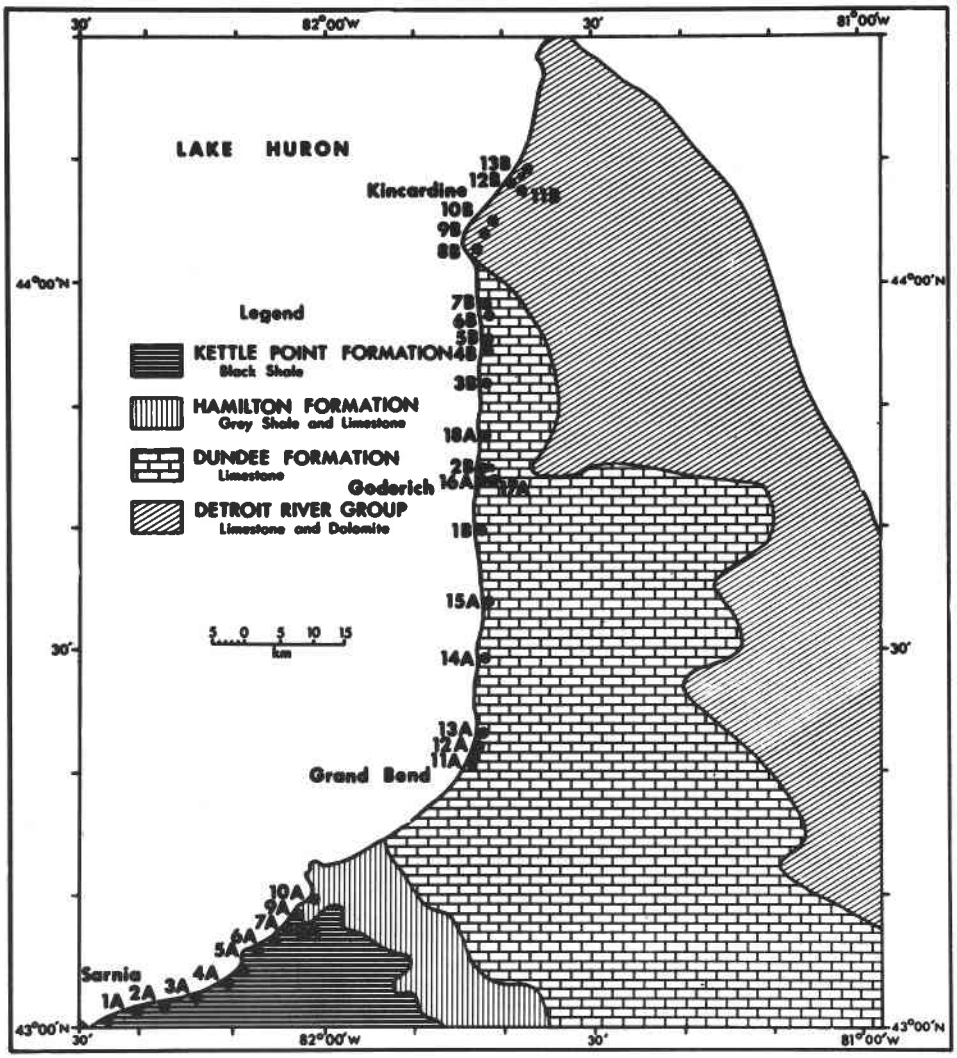


FIG. 2. Bedrock geology and sampled locations. Geology from maps of the Geological Survey of Canada.

cluster. Thus the additional sampling has shown that the "A" and "B" tills are not confined geographically as previously suspected.

The within-site variations in the remanence are small for most sites and probably reflect variations in the Earth's magnetic field directions during the short period of time during which the till was being deposited. The between-site variations in the mean site remanence direction (Table 1) probably arise from the variation in the magnetic field over a somewhat longer period—the order of a few hundred years (Gravenor *et al.* 1973).

The mean remanence directions and the VGP positions calculated from them using the dipole formula show that the "A" and "B" till sheets are well separated and have non-intersecting ovals of 95% confidence (Figs. 3 and 4). This suggests that each till sheet may be characterized by its VGP position. The VGP position for the Walkerton varved clays falls between the mean positions for the "A" and "B" tills and is close to

the position obtained by combining the data for the "A" and "B" tills into a single VGP position (Fig. 4).

Magnetic Till Fabric

The anisotropy of magnetic susceptibility of 10 to 30 specimens from each site was measured on a Schonstedt SSM-1A spinner magnetometer. A Toroid transformer bridge (Christie and Symons 1969) was used to determine the susceptibility along one axis. For each specimen the magnitude and direction of the minimum, intermediate, and maximum susceptibility directions were computed using the method of successive approximations (Nye 1960). The results are shown in Table 2.

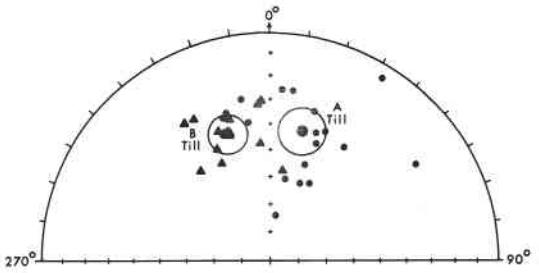


FIG. 3. Mean site remanence directions. The small black circles are the mean site remanence directions for the "A" till and the large black circle is the mean remanence direction for all the "A" sites. The small black triangles are the mean site remanence directions for the "B" till and the large black triangle is the mean remanence direction for all the "B" sites. The large open circles mark the limits of the 95% confidence level.

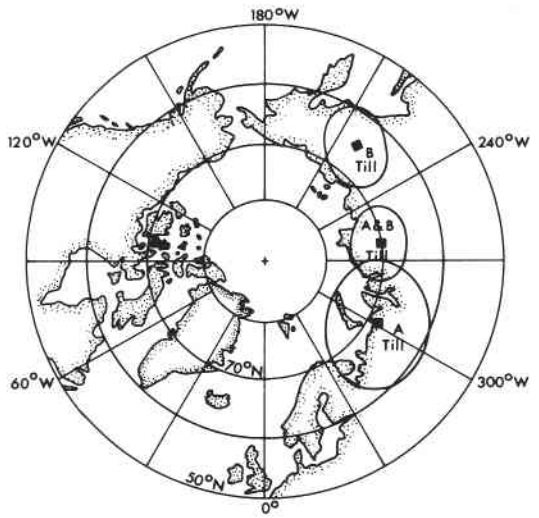


FIG. 4. Virtual geomagnetic pole positions for the "A" and "B" tills and the mean ("A" + "B") virtual geomagnetic pole position for all sites. The ovals mark the limit of the 95% confidence level.

Using the conventions for reporting the results of magnetic fabric proposed by Gravenor and Stupavsky (1975) the following observations are made:

1. The results obtained from about 5% of the specimens were rejected as they were not sufficiently anisotropic to be reliable.
2. The various sites exhibit varying degrees of fabric development. Max/Int ranges from 1.01 to 1.10 and Min/Int ranges from 0.39 to 0.97 (Table 2).
3. The site mean directions of maximum susceptibility are well defined and the 95% confidence limits about the site mean declination and mean inclination are relatively small.
4. The easterly to southeasterly directions obtained for sites 1A to 10A are at odds with other lines of evidence. For example, at site 6A a log was found embedded in the till. The long axis of the log gives a direction of 175° and dipped 10° to the north. At sites 7A and 8A the long axes of large boulders and striations on the boulders give directions of 190°. Some of the directions are normal to the long axis of the Wyoming moraine and others are parallel to the moraine (Fig. 1).

These observations suggest that there may be differences in the mode of formation of the magnetic fabric in the till and that some of the directions may be at right angles to the ice-movement direction. These 'abnormalities' may be explained in part by the possibility that the magnetic fabric at a few of the sampled sites may be transverse to the ice flow direction. Although a transverse magnetic fabric has not been previously reported, such a possibility should not be surprising, as it has been known for some time that many pebble fabric determinations show strong transverse directions (*e.g.* Harrison 1957). Figure 5 is a rose diagram showing the magnetic fabric data for site 16A. The two peaks on the rose diagram are very close to 90° apart and roughly equal in magnitude. In the absence of other evidence it is obvious that it is difficult to interpret results that show transverse peaks which are as strongly developed as those shown in Fig. 5.

The directions shown on Fig. 1 north of Grand Bend are interesting in that many of the

TABLE 2. Anisotropy of magnetic susceptibility data

Site	Number of specimens	Direction of maximum susceptibility: Declination (degrees)	Direction of maximum susceptibility: Inclination (degrees)	Direction of maximum susceptibility: α_{95} (degrees)* Declination	Direction of maximum susceptibility: α_{95} (degrees)* Inclination	Direction of minimum susceptibility: Declination (degrees)	Direction of minimum susceptibility: Inclination (degrees)	Direction of minimum susceptibility: α_{95}** (degrees)	Ratio of magnitudes: Maximum: Intermediate	Ratio of magnitudes: Minimum: Intermediate
1A	11	106.3	1.1	12.9	3.0	322.1	82.7	6.5	1.02	0.96
2A	14	91.1	5.7	7.3	5.1	203.0	78.2	9.0	1.03	0.97
3A	12	115.9	1.8	8.1	7.0	186.8	83.0	11.01	1.03	0.96
4A	13	104.8	2.5	10.0	4.3	185.8	81.1	5.7	1.03	0.94
5A	25	100.8	3.7	6.1	3.6	336.8	82.0	4.8	1.02	0.96
6A	20	105.3	1.7	10.3	4.5	164.1	82.4	5.1	1.02	0.94
7A	11	96.1	4.3	15.4	4.5	33.5	87.3	9.9	1.01	0.94
8A	13	100.7	5.5	10.8	4.3	182.2	73.3	10.0	1.02	0.95
9A	14	86.2	9.7	13.8	6.7	333.6	74.9	9.5	1.02	0.93
10A	10	114.3	1.5	5.5	3.8	192.1	79.7	5.2	1.02	0.97
11A	23	89.8	0.2	2.7	0.3	334.1	89.2	1.4	1.10	0.39
12A	16	190.2	2.3	10.8	3.8	354.6	82.6	3.9	1.01	0.92
13A	16	102.5	−2.9	10.4	5.0	42.6	82.2	4.0	1.02	0.92
14A	22	99.5	5.71	7.4	1.7	337.2	84.7	3.4	1.03	0.92
15A	11	188.4	−2.8	9.4	3.5	79.4	81.8	4.8	1.03	0.93
16A	21	199.2	−9.6	10.7	4.9	246.1	87.2	5.6	1.03	0.92
17A	11	200.0	−7.9	8.5	6.5	101.2	85.6	9.6	1.01	0.94
18A	11	183.2	−4.1	17.2	3.8	10.0	83.1	4.2	1.03	0.93
1B	18	94.6	4.0	9.5	2.4	25.2	84.0	2.6	1.01	0.92
2B	22	127.5	−6.8	10.1	1.6	68.8	73.6	2.4	1.02	0.91
3B	13	112.5	1.3	6.5	2.7	135.6	87.4	3.5	1.04	0.92
4B	12	174.3	−4.0	12.4	2.9	61.2	84.1	3.7	1.02	0.92
5B	19	113.9	−0.7	8.8	1.9	147.9	84.8	3.9	1.03	0.93
6B	17	123.0	−2.5	12.0	3.2	59.3	85.3	3.1	1.02	0.94
7B	28	158.6	−7.7	8.4	2.3	70.3	83.1	3.4	1.03	0.91
8B	15	98.5	−0.6	10.7	3.4	60.3	86.0	3.7	1.01	0.95
9B	11	102.3	−3.2	14.1	7.7	97.2	82.0	6.4	1.01	0.95
10B	24	235.1	−1.5	4.4	2.5	61.0	80.0	3.1	1.03	0.95
11B	11	229.4	−5.8	10.9	3.8	249.3	86.3	6.6	1.01	0.96
12B	10	108.4	1.0	12.1	4.9	343.5	80.8	4.7	1.01	0.96
13B	16	228.6	+2.4	9.4	3.7	190.1	78.7	4.9	1.02	0.93

*α_{95} refers to the 95% confidence limits about the mean declination and mean inclination as defined by scalar statistics.
**α_{95} is the radius of the 95% confidence circle about the mean directions as given by Fisher statistics.

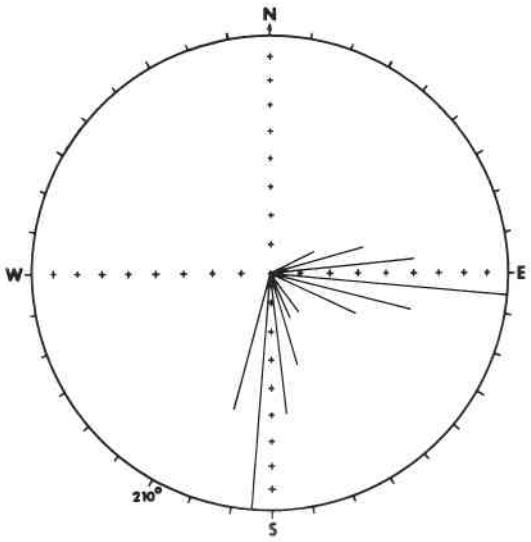


FIG. 5. Rose diagram illustrating maximum magnetic susceptibilities from samples from site 16A. The length of each line represents the number of specimens giving a specific direction.

directions are normal to the long axis of the Wyoming moraine and others are parallel to the long axis of the moraine. As mentioned under the section dealing with remanence results, there is the possibility that two till units were sampled. Therefore the magnetic fabric results were matched against the two distinct clusters of remanence results, but no correlation could be found.

As will be discussed later in this paper, some of the tills exposed along the east coast of Lake Huron may be flow till or melt-out till, and owing to the difficulties in recognizing this type of till in the field it is possible that some magnetic fabric measurements may have been made on this type of till. If this were the case, however, it would be expected that the principal directions would be randomly oriented or at least inconsistent in direction.

As there is no apparent correlation of remanence with fabric results, and it is unlikely that the variation in direction is associated with ablation till, it would appear that the most plausible explanation is that certain of the directions shown on Fig. 1 are in fact transverse directions. It is noted that if many of the directions north of Grand Bend are rotated through 90° they become normal to the long axis of the Wyoming moraine.

Physical, Chemical, and Mineralogical Characteristics and Age of the Till

Where sampled, the St. Joseph Till is grey to light brown and the average granulometric analysis of the −2 mm fraction for all 31 sites is 11.22 ± 3.1% sand, 44.8 ± 1.9% silt, and 44.57 ± 2.5% clay (average values derived from site data on Table 3). At some locations (*e.g.*, immediately south of Goderich near site 16A and west of Sarnia near site 5A, Fig. 1) where the till has been freshly exposed, it displays a laminated appearance and contains stringers and pockets of stratified sand, silt, and clay (Fig. 6). The inclusions of laminated material are usually contorted and it is quite possible that this till originated as a supraglacial deposit and is a flow till as defined by Boulton (1971). For the most part, however, the till is massive in character and probably basal in origin. It should be pointed out, however, that if the till surface is not fresh it is difficult, if not impossible, to see the laminations.

In order to determine if the "A" and "B" till units can be separated by techniques other than remanence data, the pebble lithology, particle-size distribution, carbonate content, specific magnetic susceptibility (SMS), and ratio of purple (dark red) to white (light red) garnets were determined. A minimum of 50 pebbles were identified from the +2 mm fraction of a 500 g sample of till from each site. Particle-size distribution was made by sieve analysis to determine the amount of material in the sand fraction; clay was determined by the hydrometer method and silt by difference. The calcite and dolomite contents were obtained by using a Chittick

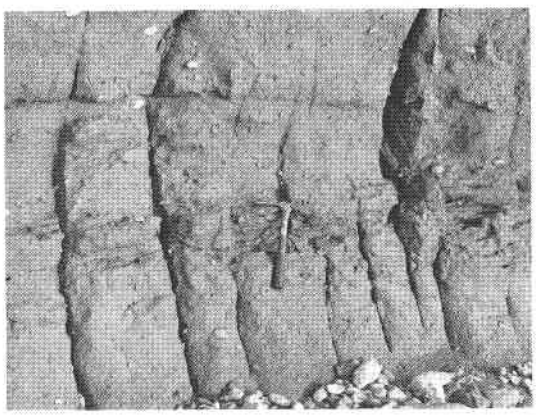


FIG. 6. Photograph of till near site 5A.

TABLE 3. Physical and lithological properties of till

Site	Granulometric analysis: Sand −2 +0.062 mm	Silt −0.062 +0.004 mm	Clay −0.004 mm	Carbonate determination: Calcite (%)	Dolomite (%)	Total carbonates (%)	Ratio calcite:dolomite	Pebble lithology: Igneous + metamorphic (%)	Shales + sandstone (%)	Carbonates (%)	Specific magnetic susceptibility: +1 mm	−1 +0.5 mm	−0.5 +0.25 mm	−0.25 +0.125 mm	−0.125 +0.062 mm	−1 mm Bulk	Bedrock geology
1A	9.7	38.9	5.14	11	18	29	0.61	16.2	30.3	53.5	+00.1	−22.4	−5.7	25.8	33.6	30.6	
2A	16.7	38.8	44.5	12	16	28	0.75	11.2	28.8	60.0	+48.0	1.5	−96.0	17.7	20.6	29.9	
3A	7.6	37.1	55.3	9	15	24	0.60	16.0	17.3	66.7	27.0	−24.2	−6.7	6.7	26.7	35.1	
4A	6.9	26.8	66.3	18	11	29	1.63	5.6	9.9	84.5	88.8	−57.2	0.0	8.4	33.6	29.9	Kettle Point
5A	12.1	43.4	44.5	14	17	31	0.80	14.2	25.2	60.6	−64.0	−70.2	−18.9	0.0	14.7	26.5	Formation
6A	16.2	46.1	37.7	15	20	35	0.75	11.4	13.3	75.3	−54.8	−58.0	0.0	9.9	53.8	35.4	Black shale
7A	13.3	42.9	43.8	15	23	38	0.65	12.0	12.0	76.0	−140.4	−80.6	−16.3	−13.7	33.9	38.6	
8A	12.5	49.1	38.4	19	12	31	1.58	12.0	25.6	62.4	−67.5	−36.3	12.3	13.0	38.4	30.0	
9A	12.8	43.7	43.5	18	18	36	1.00	16.2	9.8	74.0	−49.2	0.0	0.0	8.9	54.6	37.6	
10A	7.3	55.3	37.4	18	13	31	1.4	13.5	5.4	81.1	−145.1	−263.8	−25.0	12.7	37.2	30.2	
11A	12.2	37.0	50.8	23	16	39	1.44	11.3	2.7	86.0	−51.1	−15.8	0.0	5.2	10.2	30.0	
12A	13.3	38.7	48.0	21	21	42	1.00	11.3	5.0	83.7	50.4	47.2	20.0	26.6	66.6	28.3	
13A	14.4	41.6	44.0	20	24	44	0.83	12.8	4.6	82.1	0.1	13.1	12.3	23.9	62.3	30.5	
14A	2.7	53.3	44.0	29	24	53	1.21	7.9	1.8	90.2	−85.3	24.4	−29.6	63.9	17.7	24.0	
15A	6.6	44.7	48.7	29	21	50	1.38	6.5	0.9	92.6	−23.3	4.9	34.9	29.9	38.7	25.5	
16A	3.1	46.1	50.8	16	21	39	0.76	9.3	5.4	85.4	−90.1	13.3	−5.4	9.7	46.6	30.2	
17A	13.2	47.8	39.0	21	24	45	0.87	14.0	6.0	80.0	6.7	5.3	11.6	27.6	58.1	30.6	
18A	11.8	50.0	38.2	21	26	47	0.81	10.9	3.3	85.8	14.9	−7.0	11.8	22.1	48.9	32.3	Dundee
1B	12.1	44.2	43.7	19	24	43	0.79	11.8	8.2	79.9	316.1	0.1	12.2	22.4	51.3	31.1	Formation
2B	1.7	59.3	39.0	21	21	42	1.00	11.9	0.0	88.1	590.4	−246.5	−58.9	−10.2	10.6	31.1	Limestone
3B	4.0	38.6	57.4	20	17	37	1.18	3.0	3.0	94.0	−57.7	35.1	0.0	11.5	19.5	29.5	
4B	1.1	39.7	59.2	22	24	46	0.92	0.0	0.0	100.0	−127.6	−138.5	−59.3	−22.3	21.3	29.1	
5B	13.3	49.0	37.6	18	16	34	1.12	14.3	3.1	82.6	−16.4	1.9	15.0	22.4	23.7	33.8	
6B	17.0	44.6	38.4	22	24	46	0.92	15.8	5.4	78.8	−8.6	7.9	20.4	43.1	63.1	35.0	
7B	9.3	50.7	40.0	18	27	45	0.67	20.8	1.9	77.4	57.9	7.3	5.7	13.7	22.6	32.0	
8B	15.8	45.9	38.3	24	22	46	1.09	11.2	8.9	79.9	−61.5	−32.9	−25.9	21.0	85.2	41.5	
9B	14.0	47.0	39.0	21	26	47	0.81	15.1	8.8	76.1	−36.3	−111.6	−6.7	17.5	49.1	37.5	Detroit River Group
10B	13.3	50.0	36.7	19	25	44	0.76	9.8	0.5	89.6	−167.5	−50.0	11.8	24.1	63.7	41.5	
11B	20.7	46.1	33.2	22	23	45	0.96	15.9	3.6	80.5	31.6	−34.0	8.6	34.5	74.0	49.2	Limestone and
12B	20.0	42.7	37.3	22	22	44	1.0	5.6	0.9	93.5	26.6	0.1	0.1	39.9	76.5	48.4	dolomite
13B	15.7	42.7	41.6	21	21	42	1.0	15.0	5.9	79.1	−23.4	−69.1	0.1	21.6	68.8	34.7	
								Average and standard deviation									
1–10	11.5	42.2	46.2	14.9	16.3	31.2	0.98	12.8	17.8	69.4	−35.7	−61.1	−15.7	7.6	34.7	32.4	Kettle Point
S.D.	3.5	7.6	9.1	3.4	3.7	4.1	0.4	3.2	8.9	10.2	76.7	76.5	30.5	11.6	12.7	4.0	Formation
11–18	9.7	44.9	45.4	22.6	22.3	44.9	1.04	11.0	3.6	85.4	−22.2	10.7	6.9	26.1	43.6	28.9	
S.D.	4.8	5.6	4.9	4.5	3.1	5.4	0.26	3.0	1.9	4.3	49.8	19.3	19.1	17.6	20.5	2.8	Dundee Formation
19–25	8.2	45.9	45.9	20.0	21.9	42.2	0.95	11.1	3.1	85.7	107.7	−47.5	−9.3	11.5	30.3	31.7	Limestone
S.D.	6.8	7.5	9.9	1.7	4.1	5.1	0.15	7.1	2.9	8.3	255.0	104.5	34.6	21.8	19.2	2.1	
26–31	15.5	46.4	40.8	21.6	23.5	45.0	0.93	12.1	4.8	83.1	−38.4	−49.6	−2.0	26.4	69.5	42.1	Detroit River Group
S.D.	3.9	3.1	4.9	1.4	2.0	1.8	0.12	4.0	3.7	6.8	72.8	38.0	13.4	8.8	12.3	5.8	Limestone and dolomite

gasometric apparatus (Dreimanis 1962). The garnet ratios were measured by examining the heavy minerals from the −60 + 230 mesh size fractions. The specific magnetic susceptibility of the −2 mm fraction of the till was measured on a toroid transformer bridge (Christie and Symons 1969).

It is evident from Fig. 2 that the till comprising the "A" and "B" types overlies different bedrock lithologies. Therefore, instead of entering all data into a discriminant analysis it was decided to break the data into three units (Table 3) based on bedrock lithology. The reason for breaking up the data into these units is readily evident from an examination of Table 3. For example, the till from sites 1A to 10A, which overlies the black shales of the Kettle Point Formation, has significantly smaller amounts of total carbonates both in the matrix and pebble fractions and significantly larger amounts of shale and sandstone pebbles. Till from sites 8B to 13B, which overlie the limestones and dolomite of the Detroit River Group, has a higher sand and smaller clay content and significantly higher bulk susceptibility and specific magnetic susceptibility (SMS) in the −0.125 mm + 0.62 mm size fraction than the other three units. When the tills are broken into three units 1A to 10A, 11A to 18A plus 1B to 7B, and 8B to 13B (Fig. 2), and the data from Table 3 entered into a BMD 07M discriminant analysis, a perfect separation is obtained. In fact the perfect separation can be obtained after only a small number of variables have been entered into the discriminant function. It should be noted that when the SMS of the six size ranges is entered alone in the discriminant analysis a good separation is obtained between the "A" and "B" till units, the "B" till having a significantly higher bulk susceptibility and SMS than the "A" till. As pointed out by Gravenor and Stupavsky (1974) the magnetic material in till in southern Ontario is derived primarily from the Canadian Shield and decreases approximately exponentially with distance south of the Canadian Shield–Paleozoic boundary. Therefore because the "B" till is closer to the Shield than the "A" till it is to be expected that an analysis of the SMS measurements would demonstrate the presence of two tills because of the 'end-member' effect.

It is clear, therefore, that the chemical and physical characteristics of the upper till are highly dependent upon the nature of the underlying bedrock and the distance from the Shield. Hence these variables are unsuitable for the purpose of separating the till units in the area under study. This is not to suggest that these techniques are not useful in other situations, but rather to point out that the use of these methods and the statistical manipulation of the results must be used with caution as 'end-member' effects can distort the results.

In order to get around this problem it was decided to examine the heavy minerals from 15 of the sites (11A to 18A and 1B to 7B) with particular reference to the ratio of purple (dark red) to white (light red) garnets in the till. As the transluscent heavy minerals found in till have been derived primarily from the Shield (Gravenor 1951), variations in heavy minerals in tills reflect differences in the source area (Gwyn 1971). The use of garnet ratios to differentiate tills in southern Ontario was first used successfully by Dreimanis and Reavely (1953), and because garnets are readily identifiable this technique presents a reasonably fast method to determine if there are variations in source for the "A" and "B" tills. The heavy minerals were separated from the −60 + 230 mesh fraction using tetrabromoethane (SG 2.96) as the separation medium. Some 200 transluscent minerals were identified from each sample. As garnets comprise from 15 to 20% of each sample, the number of garnets identified from each sample is in the range of 30 to 40. The results of this analysis are shown in Table 4.

Other heavy minerals identified in descending order of abundance are amphiboles (primarily hornblende), ortho- and clinopyroxenes, epidite, zircon, sphene, rutile, zoisite, biotite, and tourmaline.

The results show that the "B" till contains more garnet than the "A" till and a 't' test shows that the mean ratio of purple to white garnets of the "B" till is significantly different from the "A" till at the 90% level of confidence. When the percentage of all heavy minerals is entered in a BMD 07M discriminant analysis only one site in 15 is misclassified. Using only the percentage of purple and white garnets and the ratio of purple to white garnets in the discriminant analysis, 5 sites of 15, or 33% of the sites, are misclassified; but the group means are significantly different at the 95% level of confidence. It is considered that this information supports the possibility that two different tills were sampled along the east coast of Lake Huron.

TABLE 4. Analysis of garnet content of the "A" and "B" tills

Colour of garnet and ratios of purple to white garnets	"A" till		"B" till	
	Mean (%)	S.D.	Mean (%)	S.D.
Purple (dark red)	9.45	3.68	12.64	4.34
White (light red)	4.88	2.44	8.37	2.44
Purple/white	2.31	0.96	1.62	0.56

As mentioned under the section on magnetic fabric, a log was found embedded in the St. Joseph Till near site 6A. The precise location of the find is lat. 42° 03′ 08″ N, and long. 82° 09′ 08″ W. The log is about 12 cm in diameter and about 1 m in length and has been identified as *Picea* sp. (spruce) by L. D. Farlay-Gill, Geological Survey of Canada. It was found in basal till about 3 m above water level, and a layer of silt a few millimetres thick around the outside of the log separated the log from the enclosing till. This log has been dated by the radiocarbon method at 13 100 ± 110 y BP (GSC-2213). This data is compatible with estimates of age of the Port Huron stadial (Dreimanis and Karrow 1972).

Conclusions

The existence of two till units within the St. Joseph Till exposed on the east shore of Lake Huron is demonstrated by paleomagnetic evidence and differences in the ratio of purple to white garnets. The mean "A" and "B" till remanence directions are well separated with non-intersecting ovals of 95% confidence. This in itself suggests a difference in the time of deposition within the St. Joseph Till and hence the possibility of two till units in the area. The ratio of purple to white garnets for the "A" and "B" tills is significantly different. As this ratio provides information on the source area for the tills on the Shield, the difference in the ratio suggests that the "A" and "B" units had different source areas, which supports the concept of two tills.

Other physical and chemical properties such as granulometric analysis, pebble analysis, and carbonate content show variations that are directly related to the nature of the underlying bedrock and are not useful in differentiating the "A" from the "B" till. The ice movement direction determined from an analysis of the magnetic fabric is difficult to interpret as some of the directions are normal to the long axes of the moraines. It is possible that some of these directions are transverse to the ice flow, but it is also recognized that parts of the upper clay till may be a flow till and at present the nature of magnetic fabric in flow tills is not known. As there is no apparent correlation between the magnetic directions and the paleomagnetic data it is evident that this technique cannot be used to demonstrate the presence of two tills.

Aside from the variation in garnet ratios, it is evident that the only positive evidence that indicates the existence of two upper clay tills is the presence of two clusters of remanence directions. At this stage it must be kept in mind that this information only demonstrates that there was a shift in the VGP position during the time when the till(s) was being deposited in the Huron basin. As there are several sub-parallel moraines making up the Port Huron system there is always the temptation to associate the tills that lie 'inside' the moraines with the moraines themselves and think in terms of the till behind the moraines as being deposited at the same time or shortly after the construction of the moraine. In this regard Dreimanis and Morgan (1975) have related the construction of the Seaforth moraine to the Port Bruce Stadial and the Wyoming moraine to the Port Huron Stadial. Their studies indicate the presence of two tills associated with the construction of these moraines, separated in some places by Mackinaw Interstadial lacustrine or littoral deposits related to Lake Arkona. Thus it cannot be discounted that the two units recognized within the St. Joseph Till may in fact be related to two very distinct ice advances.

It must be kept in mind, however, that the interpretation placed on the development of the Seaforth and Wyoming moraines by Dreimanis and Morgan (1975) is based upon inland observations and, to our knowledge, no consistent break representing the Mackinaw Interstadial deposits has been found in the St. Joseph Till along the coast of Lake Huron.

As an alternative explanation, it is quite

possible that part of the St. Joseph Till was deposited as a basal till by the advancing ice and later covered by till that was deposited by ice retreat. Under these conditions of continuous ice cover it would be very difficult, if not impossible, to differentiate these deposits of differing age in the field.

This alternate possibility could explain the time spread as shown by the paleomagnetic data and also explain the fact that portions of the "A" till are found in "B" localities and vice versa. It is finally of interest to note that the VGP position for the Walkerton varves falls very close to the combined position for the "A" and "B" tills.

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Note

After completing this study the authors received a letter from A. J. Cooper, who has been involved in the mapping of the surficial deposits of the area south of Grand Bend. In this letter Cooper states that on the basis of laboratory analysis he has been able to distinguish two units within the St. Joseph Till, but that the two units are inseparable in the field on a consistent basis. Cooper goes on to state that it would appear that one unit predates Lake Arkona and the second postdates Lake Arkona. This information provided by Cooper supports the hypothesis that two till units were sampled along the coast of Lake Huron.

BOULTON, G. S. 1971. Till genesis and fabric in Svalbard, Spitsbergen. *In* Till: A Symposium (R. P. Goldthwait (*Ed.*)). Ohio State Univ. Press, pp. 41–72.

CHRISTIE, K. W. and SYMONS, D. T. A. 1969. Apparatus for measuring magnetic susceptibility and its anisotropy. Geol. Surv. Can., Pap. 96–41, 10 p.

COOPER, A. J. and CLUE, J. 1974. Quaternary geology of the Grand Bend area, southern Ontario. Ont. Div. Mines Prelim. Map 974.

COWAN, W. R., KARROW, P. F., COOPER, A. J., and MORGAN, A. V. 1975. Late Quaternary stratigraphy of the Waterloo–Lake Huron area, southwestern Ontario. Geol. Assoc. Can., Field Trips Guidebook, pp. 180–222.

DREIMANIS, A. 1961. Tills of southern Ontario. *In* Soils in Canada (R. F. Leggett (*Ed.*)). R. Soc. Can. Spec. Publ. 3, Univ. Toronto Press, pp. 80–96.

——— 1962. Quantitative gasometric determination of calcite and dolomite by using Chittick apparatus. J. Sed. Petrol., **32**, pp. 520–529.

DREIMANIS, A. and GOLDTHWAIT, R. P. 1973. Wisconsin glaciation in the Huron, Erie, and Ontario lobes. Geol. Soc. Am., Mem. 136, pp. 71–106.

DREIMANIS, A. and MORGAN, A. V. 1975. Differentiation of two Late Wisconsin clayey silt tills between the Seaforth moraine and Lake Huron, Ontario. Geol. Soc. Am. Abstract with Programs, **7**(6), p. 748.

DREIMANIS, A. and KARROW, P. F. 1972. Glacial history of the Great Lakes–St. Lawrence region, the classification of the Wisconsin(an) stage and its correlatives. Proc. 24th Geol. Congr. Sect. 12, pp. 5–15.

DREIMANIS, A. and REAVELY, G. H. 1953. Differentiation of the lower and upper till along the north shore of Lake Erie. J. Sed. Petrol. **23**(4), pp. 238–259.

FISHER, R. A. 1953. Dispersion on a sphere. Proc. R. Soc. (London), Ser. A, **217**, pp. 295–305.

GRAVENOR, C. P. 1951. Bedrock sources of till in southwestern Ontario. Am. J. Sci., **249**, pp. 66–71.

GRAVENOR, C. P., STUPAVSKY, M., and SYMONS, D. T. A. 1973. Paleomagnetism and its relationship to till deposition. Can. J. Earth Sci., **10**(7), pp. 1068–1078.

GRAVENOR, C. P. and STUPAVSKY, M. 1974. Magnetic susceptibility of the surface tills of southern Ontario. Can. J. Earth Sci., **11**(5), pp. 658–663.

——— 1975. Convention for reporting magnetic anisotropy of till. Can. J. Earth Sci., **12**(6), pp. 1063–1069.

GWYN, H. 1971. Heavy mineral assemblages in tills and their use in distinguishing glacial lobes in the Great Lakes region. Unpubl. Ph.D. thesis, Univ. Western Ontario, London, Ontario.

HARRISON, P. W. 1957. A clay-till fabric: its character and origin. J. Geol., **65**, pp. 275–308.

KARROW, P. F. 1974. Till stratigraphy in parts of southwestern Ontario. Geol. Soc. Am. Bull., **85**, pp. 761–768.

LEVERETT, F. and TAYLOR, F. B. 1915. Pleistocene of Indiana and Michigan and the history of the Great Lakes. U.S. Geol. Surv., Mon. 53, 816 p.

NYE, J. F. 1960. Physical properties of crystals. Clarendon Press, Oxford, 322 p.

STUPAVSKY, M., SYMONS, D. T. A., and GRAVENOR, C. P. 1974*a*. Paleomagnetism of the Port Stanley Till, Ontario. Geol. Soc. Am. Bull., **85**, pp. 141–144.

STUPAVSKY, M., GRAVENOR, C. P., and SYMONS, D. T. A. 1974*b*. Paleomagnetism and magnetic fabric of the Leaside and Sunnybrook Tills near Toronto, Ontario. Geol. Soc. Am. Bull., **85**, pp. 1233–1236.

SYMONS, D. T. A. and STUPAVSKY, M. 1974. A rational paleomagnetic stability index. J. Geophys. Res., **79**(11), pp. 1718–1720.